

BCS Practitioner Certificate in Requirements Engineering

Sample Paper B

Record your surname / last / family name and initials on the answer sheet.

This sample paper has only 40 questions, which are a mixture of multiple-choice and multiple-response questions – 1 mark awarded to each question.

Multiple choice questions allow only one correct answer to be selected for 1 mark.

Multiple response questions require the candidate to select a number of correct responses for 1 mark.

There are no trick questions.

A number of possible answers are given for each question, indicated by either **A B C D or E**. Your answers should be clearly indicated on the answer sheet.

Pass mark is 26/40

Time allowed: 60 minutes.

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This professional certification is not regulated by the following United Kingdom Regulators
- Ofqual, Qualifications in Wales, CCEA or SQA

- 1 A new business analyst (BA) is working on an Agile project. They are curious as to why the owner of each requirement is invited to a walkthrough of the requirement with the development team.

What part of the requirements engineering framework **WOULD** this activity be intended to support?

- A Requirements validation.
- B Requirements analysis.
- C Requirements documentation.
- D Requirements management.

- 2 The following is taken from a project initiation document (PID);
"The project must be implemented using the corporate approved change framework. It must pass all audits against the framework and must use all mandatory and advised documents of the framework."

Which section of the PID **SHOULD** this appear in?

- A Authority.
- B Scope.
- C Constraints.
- D Resources.

- 3 Which of the following is an example of tacit knowledge?

- A The documented way a task should be performed.
- B Individual's job description.
- C Corporate culture.
- D Branding style guide.

- 4 Which of these elicitation techniques would be suitable for establishing tacit knowledge?

Select **two** CORRECT responses.

- A Interviews.
- B Scenario analysis.
- C Document analysis.
- D Questionnaire.
- E Shadowing.

- 5 A BA is working on a project where three stakeholders have differing views on the best customer payment process to use in their project, leading to frustration and disengagement. The BA has decided to hold a workshop.

Which of the following **BEST** describes the BA's motivation for holding this workshop?

Select **two** CORRECT responses.

- A Workshops are always more cost effective than individual interviews.
- B Workshops can help stakeholders to understand one another's perspectives.
- C Workshops are the most suitable method for identifying business rules.
- D Workshops are the only means by which stakeholders can communicate their requirements.
- E Workshops can help stakeholders feel involved and committed.

- 6 A team of business analysts are looking to elicit some requirements. Their manager has suggested scenario analysis as a great way of eliciting complex requirements.

Which of the following uses of scenario analysis **SHOULD** the manager use to justify their choice?

Select **two** CORRECT responses.

- A Scenario analysis can be used to develop test cases and acceptance criteria.
 - B Scenario analysis can be used to cross check processes where some extensions may have been missed.
 - C Scenario analysis is always undertaken in a simulated environment, so has no impact on real business activity.
 - D Personas will be built from the scenario analysis to help illustrate the benefits of the changes.
 - E Scenario analysis focuses on eliciting non-functional requirements.
- 7 A business analyst is looking for a way to elicit requirements for a new internal website for a company. The site will be used to reveal and hold the company values and strategy. Stakeholders are struggling to articulate their requirements because they keep trying to define the solution.

Which of the following elicitation techniques **SHOULD** be used to help stakeholders visualise the solution, to elicit requirements?

- A Prototyping.
- B Workshops.
- C Scenario analysis.
- D Document analysis.

8 Which of the following **WOULD** you look for when choosing an elicitation technique to use in an Agile project environment?

- A Significant time taken to perform the technique.
- B Ease of access to documentation.
- C Ease of repeatability.
- D High volume of stakeholders involved.

9 A business analyst has been asked to explain the reasons for documenting requirements to a group of peers.

Which of the following are benefits of documenting requirements?

Select **three** CORRECT responses.

- A Ensures correct prioritisation.
- B Enabling communication.
- C Provides a basis for validation.
- D Ensures consistency.
- E Clarifies ownership.

10 Which of the following documentation types **COULD** be classed as diagrammatic?

Select **three** CORRECT responses.

- A Business process.
- B Use case diagram.
- C User story.
- D Requirements catalogue.
- E Data model.

- 11 Which characteristic of a requirements catalogue entry would **NOT** be completed during requirements elicitation?
- A Description.
 - B Author.
 - C Business Area.
 - D Priority.

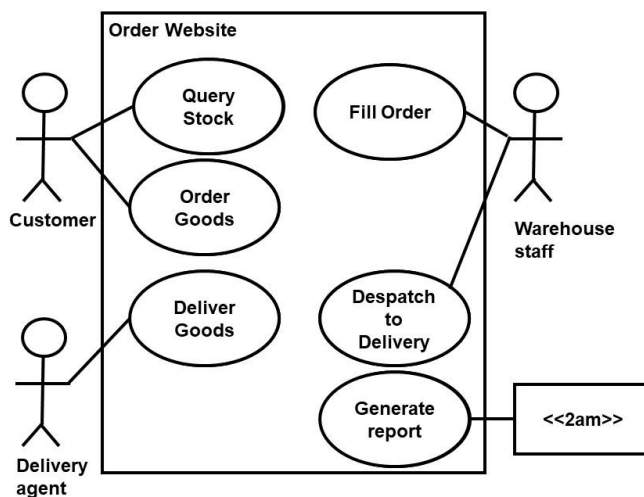
- 12 An employee has written the following user story: "As a user I want to capture the customer's name and date of birth so that I can validate them as a customer next time they contact us".

In a peer review a colleague has raised that one of the elements of this story is "not as useful as it could be".

Which of the following **BEST** illustrates their reasoning?

- A The user is not clearly defined, it should refer to the perspective of a single user role.
 - B The requested feature is not explicit enough and should include what data is required.
 - C The user story does not include enough information, as there is no clear goal.
 - D The goal of the user story is too explicit and there are other reasons why this data may be needed, and it should not be constrained to just this purpose.
- 13 Which of the following statements **CORRECTLY** explains the purposes of modelling requirements during a project?
- Select **three** **CORRECT** responses.
- A Clarifying understanding.
 - B Generating questions.
 - C Defining business rules.
 - D Prioritising changes.
 - E Escalating issues found to senior leaders.

- 14** Select the CORRECT word to complete the following description for the rationale for modelling requirements during requirements engineering.
"We need models to communicate and describe the _____ and data requirements for our system."
- A** Management.
 - B** Standards.
 - C** Functionality.
 - D** System.
- 15** A business analyst has created this use case diagram for an upgrade to the online retail system.



A stakeholder has pointed out that on occasions, customers still call a sales team to place orders because there is a sales phone number on the website.

What action should the business analyst take with this new information?

- A** Nothing on this diagram, but make sure that the "Call Sales" process and requirements are unchanged by this project.
- B** Add a new actor for "Sales Agent" and associate them with the "Fill Order" use case.
- C** Add a use case for "Phone Orders" with an association to "Customer" and a new actor for "Sales Agent".
- D** Add a use case for "Phone Orders" and associate it only with "Customer" because "Sales Agents" are not a stakeholder on this project.

- 16** A new business analyst has joined a team. Previous team members have created use case diagrams to help explain the project.

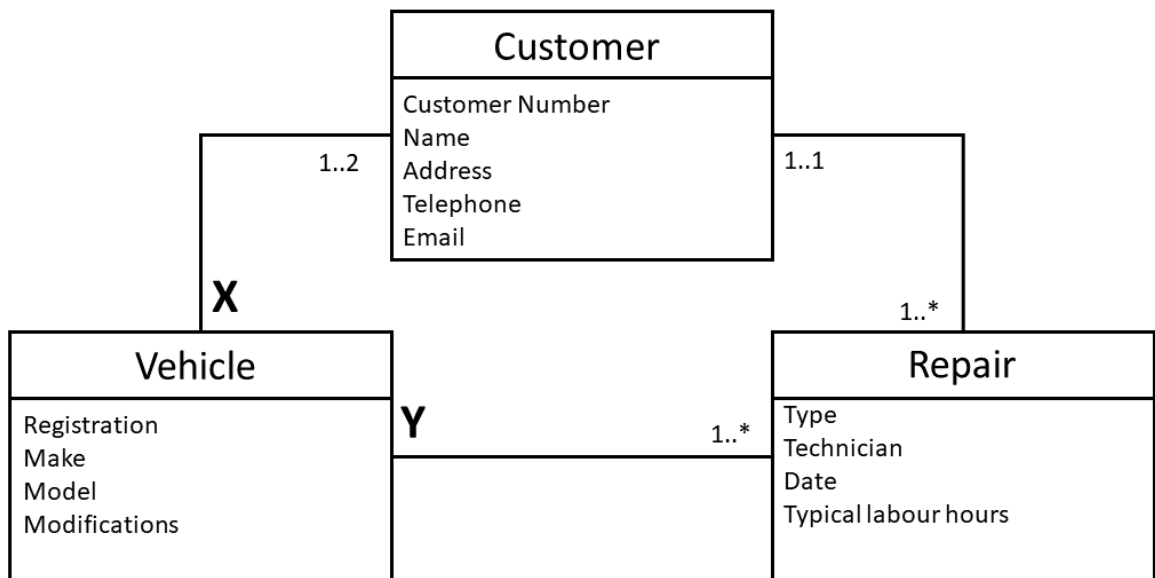
The Business analyst is curious as to why there are no actors or events shown within the system boundary.

Why **SHOULD** this be the case?

- A** They have been missed and should be added.
- B** Actors are only shown that are external to the system and interface with it.
- C** The actors that only operate within the system are not relevant to how the system works.
- D** The use cases have not been updated recently and missed the new information.

17 A business analyst is working at a car repair garage, and they are in the early stages of preparing a class diagram, as shown below. They have been given the following information:

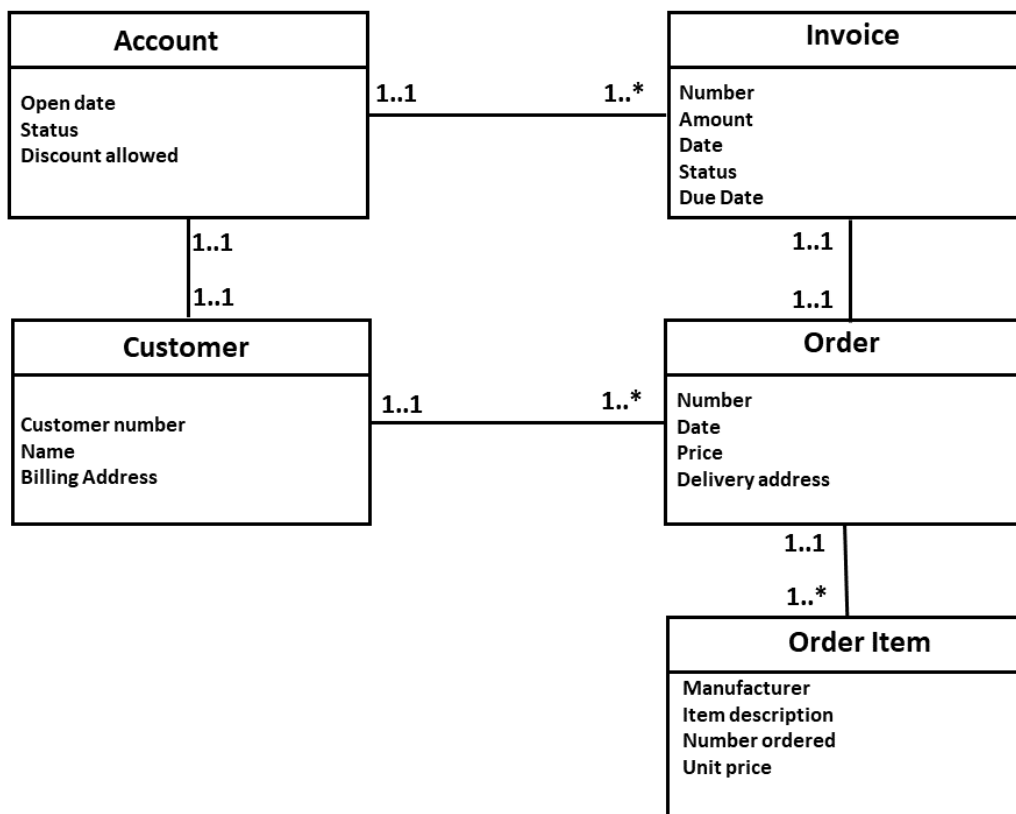
- Each customer must be associated with at least one vehicle, although they may own many.
- Every vehicle must be associated with at least one customer, although a maximum of two customers can be on record to allow for car sharing.
- Every repair may be associated with only one vehicle, although a vehicle may require multiple repairs.



What are the CORRECT multiplicities for the missing fields X and Y?

- A** X is 1..* Y is 1..1
- B** X is 1..* Y is 0..1
- C** X is 0..* Y is 0..2
- D** X is 1..* Y is 1..*

- 18** The following class diagram has been created to support the creation of a new application for customer ordering and accounts.
The stakeholder responsible for invoices is concerned that on this diagram, every order currently needs an invoice, but the correct process is to issue one invoice each month for all the orders.



What **SHOULD** the analyst do to allow for this scenario?

- A** Add an attribute to order so that payments can be recorded against the order.
- B** Make the relationship between invoice and order 1..* so that more than one order can be on each invoice.
- C** Make the relationship between account and customer 1..* so that a customer can have more accounts.
- D** Add an attribute to Invoice to hold the number of orders on each invoice.

- 19 What is the purpose of a CRUD matrix?
- A To provide details of use cases and their implementation.
 - B To illustrate who is involved in creating documentation.
 - C To identify the attributes of classes within a project.
 - D To provide a cross reference between the items in two related models.

- 20 In the following extract from a CRUD Matrix, which models are being cross checked?

	Customer	Invoice	Order	Order Item
Create New Customer	C			
Place Order	R	C	C	C
Pay Invoice		U		
Fufil Order			U	U
Deliver Order			U	U
Cancel Order	R	D	D	D

- A Use cases and classes.
 - B Actors and use cases.
 - C Actors and classes.
 - D Use cases and attributes.
- 21 A business analyst has made the decision to use prototyping to help elaborate requirements with their stakeholders.

Which of the following is **LEAST** likely to be achieved through prototyping?

- A Increased understanding of the requirements by stakeholders.
- B Visual representation of requirements which can be shared with stakeholders.
- C Rejection of requirements.
- D Verification of requirements.

- 22** An employee has been assigned to a project and is looking through their role terms of reference. It says that they will be providing specialist industry knowledge and will be working with others to help them gain a greater depth of understanding of the requirements.

Which role has the employee been allocated to?

- A** Tester.
- B** Product Owner.
- C** Subject Matter Expert.
- D** Business analyst.

- 23** Which of the following statements apply to the rationale and purpose of validating requirements?

Select **two** CORRECT responses.

- A** Making sure that the requirements are aligned to the business objectives.
- B** Making sure that the accountable person confirms that the requirement(s) are correct and can be built or enter the backlog.
- C** Making sure that the requirements meet the solution design.
- D** Making sure that the right method has been followed to get the requirements ready.
- E** Making sure that the estimate for delivering the change is reasonable and can be met.

- 24** Which of the following is **NOT** an objective of the analysis stage of the requirements engineering framework?

- A** Ensuring requirements developed clearly.
- B** Ensuring that they comply with standards.
- C** Ensuring the validation of requirements.
- D** Ensuring requirements are well organised.

- 25 A business analyst is prioritising stories in a backlog. They have a requirement that has been raised by a subject matter expert, to make the solution able to provide information about the number of transactions being handled and which teams people are allocated to.

It is not specifically required by the business case, but it is an excellent idea.

Which MoSCoW rating is **MOST LIKELY** to be applied during initial backlog reviews?

- A Want to have, but won't this time.
- B Should have.
- C Must have.
- D Could have.

- 26 A business analyst has received a requirement to create customer accounts in a purchasing system. This feature does need to be present in the solution and the capability is well understood, but there is scope to use a "guest" function for customers until this can be delivered.

Which MoSCoW rating is **MOST LIKELY** to apply to this requirement ?

- A Should have.
- B Must have.
- C Could have.
- D Want to have, but won't this time.

- 27 A business analyst has discovered a pair of duplicate requirements during analysis. After ensuring that all aspects of both requirements are captured in a single requirement, the duplicate will be closed.

Which of the following attributes **CANNOT** be merged into a single requirement, and therefore must be changed?

- A Owner.
- B Description.
- C Acceptance criteria.
- D Source.

28 When requirements are 'sliced', what purpose is being served?

- A** Making sure that each requirement is atomic.
- B** Splitting a set of requirements into sub-sets for iterative delivery.
- C** Ensuring that duplicate requirements are removed.
- D** Ensuring that the greatest value items are delivered first.

29 Business rules will be found as the requirements are expressed and elaborated, through the production of models throughout requirements engineering.

Which of the following is **NOT** a place where business rules are defined?

- A** In the CRUD Matrix.
- B** In the definition of which actors perform which tasks in a process model.
- C** In the prioritisation of requirements catalogue.
- D** In the steps in a task description.

30 A business analyst has identified a requirement which they are unable to confirm as being met or not, as there is no defined criteria to check against.

What characteristic of a requirement has failed in this instance?

- A** Testable.
- B** Quality.
- C** Negotiable.
- D** Estimatable.

31 A business analyst intends to use user role analysis.

What purpose does this serve?

- A** Identification of stakeholders that will own the requirements.
- B** Designing solutions that meet industry and corporate standards.
- C** Developing an understanding of how “users” engage with a business area or service.
- D** Making sure that the user experience of using the solution meets requirements.

32 Which of the following statements describe requirements traceability?

Select **three** CORRECT responses.

- A** Evidence of how a requirement delivers a part of the business objectives.
- B** Evidence of how a requirement was created (its origin).
- C** Evidence of how a requirement was validated.
- D** Evidence of how a requirement was resolved (delivered).
- E** Evidence of how the owner of a requirement has changed over the delivery.

33 Requirements management contains six elements. Three of these are:

- Requirements cross referencing
- Software support
- Requirements identification

Select the other **three** CORRECT elements.

- A** Requirements documentation.
- B** Change control.
- C** Configuration management.
- D** Requirements origin and ownership.
- E** Requirements validity.

- 34 Which of the following gives the **BEST** rationale for the entire requirements management stage of the requirements engineering lifecycle?
- A Ensuring that the elaboration of requirements delivers requirements of the greatest value first to the development team.
 - B Ensuring that the requirements are being delivered to the project plan and will allow the project to meet its objectives.
 - C Reviewing the requirements to remove duplication or error, negotiate conflicts, evaluate feasibility and prioritise.
 - D Dealing with changes to the defined requirements and ensuring that the desired level of traceability is achieved.

- 35 A business analyst has been asked to develop a suitable change control process for a project.

What **SHOULD** be the first stage of the process?

- A Document the proposed change.
 - B Analyse the impact of the proposed change.
 - C Ensure that the change is required.
 - D Check with stakeholders that the change is owned.
- 36 Version control has been defined for a project you are working on.
- The standard is that each time a requirement is validated, it will increase the version number by 1.
 - Each time it is updated for draft purposes it will be updated 0.1 to ensure that the latest version is always available for change.

The document you are working on is version 3.4. What does this mean?

- A The document has passed validation three times and there have been four saved updates since it last passed validation.
- B The document has been updated four times and the first three have passed validation, the fourth has not yet.
- C The document has passed validation four time and there have been three updates since the last validation.
- D The document has passed validation three times and has been proposed for validation four more times and has not yet passed.

37 Validation of requirements in an Agile project are carried out at which points in their delivery?

Select **two** CORRECT responses.

- A During regular team meetings while delivering the items.
- B During a daily standup.
- C During a release retrospective when items are complete.
- D During approving items ready for development by the team.
- E During project initiation when the product backlog is produced.

38 An employee has questioned how they can record the emotional state of the persona in a customer journey map, as they are unsure if the customer is using the facility to leave good comments or complain.

Which section of the customer journey map **SHOULD** a colleague point to, to answer this query?

- A Touchpoints.
- B Stages of customer journey.
- C Persona goal.
- D Potential opportunities for improvement.

39 When planning the approach to requirements engineering for a project, which of the following must be established before moving forward?

- A The development approach to be used in the project.
- B The expected interface of the final solution.
- C Alternative options for a solution.
- D The planned time off for project team members.

- 40 A Business analyst is working on a new application to be used internally to process mortgage applications. One of the requirements reads "The daily rate of interest and the monthly repayment must be calculated".

When they are applying quality criteria, where **SHOULD** this requirement fail?

- A The requirement is ambiguous.
- B This is a statement of solution not a requirement.
- C The language used is too vague and not clear.
- D It contains multiple requirements.

End of Paper

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Sample Paper B - Answer Sheet

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Question	Answer(s)	Learning Outcome	Rationale
1	A	Topic 1 (1.2)	At the walkthrough of each requirement the objective would be to ensure that the requirement is not only correctly defined according to the rules for the team (verified) but also that the owner agrees that the delivery of the requirement will fulfil the business objective that is assigned to it. This is validation.
2	C	Topic 1 (1.4)	This element is a constraint. It is a limitation to how the project may act that is applied by the organisation itself. It is not limiting what areas of the organisation the project can operate or deliver, allocating any resources, or providing or using any authority.
3	C	Topic 2 (2.1)	Corporate culture is tacit knowledge. This is a set of rules or guidance that is not written down and is 'just known to be true' by those who use it. Corporate culture will evolve with the organisations and cannot be trained or predicted from outside the organisation.
4	B and E	Topic 2 (2.2)	Shadowing and scenario analysis both give an opportunity to engage with stakeholders about how the task is being done and why decisions and choices are being made. Then when a piece of tacit knowledge is uncovered it can be explored and documented.
5	B and E	Topic 2 (2.3)	The analyst here is trying to get the stakeholders to see each other's perspectives and come up with a way of describing the various issues in a way that allows all stakeholders to see why the alternative view exists. They are not trying to save time, or look for shortcuts. They are looking to understand why the disagreement exists around payments and get an understanding of how to approach the transactions while keeping the stakeholders engaged and onside.
6	A and B	Topic 2 (2.3)	Scenarios can be used to create test cases and to explore the extensions that may have been missed. There is no intent to capture how often or how many of each scenario occurs, and they cannot be established without knowledge of what is really happening.

7	A	Topic 2 (2.4)	Prototyping would give a really good basis for eliciting requirements for this as it would allow the stakeholders to better visualise the solution. The outcome is a set of web pages, and this could be created as a demonstration where staff could have a look and make suggestions for changes. It could also evolve into the eventual solutions and also could be used for comparison testing where various versions could be used to see which is liked the most. Workshops will be difficult and expensive to organise, scenario analysis will be difficult as this is an information site so use cases will be fewer. Document analysis could be used for some elements of the creation of the prototype, but will not elicit requirements for the stakeholders.
8	C	Topic 2 (2.5)	During an agile project you will be iterating the requirements elicitation, so the technique will need to be repeatable. The other factors are time and resource heavy, which is more likely to suit a linear development style.
9	B, C and D	Topic 3 (3.2)	Documentation is specifically done to provide consistency to the recording, communication across the group of people that need to see them, validation of a known version of the requirements and for development of a solution to the requirements. Prioritisation and ownership are valuable attributes of the requirements but are not a reason to document the requirements.
10	A, B and E	Topic 3 (3.3)	The data model, use case diagram and business process model can all be considered diagrammatic. The others are text based.
11	D	Topic 3 (3.4)	The priority would not be known at initial identification of the requirement, so the catalogue entry would not be filled. The author and business area must be known and the description will be filled with what is initially known and be updated during the elaboration of the requirement.
12	A	Topic 3 (3.5)	This is true, the list of agreed roles should be used. This user story will belong to a type of "user of the system" and not all of them will need to do this as they do not need the goal.
13	A, B and C	Topic 4 (4.2)	The purpose of modelling includes clarifying understanding, generating questions and defining business rules. Although it may highlight issues, it is not specifically for escalation of them for resolution or prioritising any of the proposed changes.

14	C	Topic 4 (4.1)	The word missing is functionality. Standards are important, but not what you are modelling, management and system descriptions will describe the environment and solution respectively, but are not part of requirements engineering.
15	A	Topic 4 (4.3a)	This diagram is only concerned with the "order website" so while the information is really valuable to the overall customer interactions with the organisation, it will not directly affect the phone orders processes. To take any of the actions listed would imply the scope of the order website includes phone activity.
16	B	Topic 4 (4.3b)	Use cases represented on a use case diagram are there to describe what the system does, not how. They show how value is delivered to the external stakeholders by the system, not what activity is taking place within the system to do it at this level. The detail will be present within any use case descriptions created or other lower level models.
17	A	Topic 4 (4.4a)	X is 1..* as each customer must be associated with a minimum of one vehicle, and they may also be associated with multiple vehicles. Y is 1..1 because each repair can only be associated with a single vehicle.
18	B	Topic 4 (4.4b)	By making the relationship 1..* It will be allowable for one invoice to apply to many orders. The effect of this will allow a monthly invoice to be prepared containing all the orders for the customer.
19	D	Topic 4 (4.5a)	A crud matrix can be used to cross check two related models. This could be classes and use cases, or actors and use cases, or process models and use cases or actors. It is there to identify how things are created, read, updated and deleted. The objective that a full lifecycle is available for each item.
20	A	Topic 4 (4.5b)	The diagram shows classes being cross checked with use cases. The class names are things and singular. The use cases are verb-noun constructs.
21	C	Topic 4 (4.6)	Prototyping is a useful tool in aiding stakeholder understanding of a requirement. The visual representation of requirements often leads to improved engagement and acceptance.

22	C	Topic 5 (5.1)	These are the expectations of a subject matter expert. They will not be accountable for decision making over priority or need, and they also will not be analysing the requirements. They may be writing some down as part of the team though.
23	A and B	Topic 5 (5.3)	The validation itself is making sure that the business objective is known and will be met, and making sure that the accountable person agrees and 'signs-off' the requirement. The other statements are useful, relevant and helpful, but they are about verifying that the requirement is to standard for the project than valid for the organisation.
24	C	Topic 6 (6.1)	The application of requirements filters are to ensure that the requirements are fit for purpose and will deliver the desired output for the organisation. Validation will follow the analysis stage.
25	D	Topic 6 (6.2)	This is not in the business case and is therefore not critical to delivering the benefit for the project, so it cannot be must or should. It has the potential to improve the outcome of the project and add to the business case, so should be considered. It may move from this category at some point due to a change in focus or better alternatives being proposed.
26	A	Topic 6 (6.2)	This requirement is a "should". It is needed for the full solution and is understood, so is mandatory. The ability to work around the lack of the capability means that it is not essential in the first iteration.
27	A	Topic 6 (6.3)	There can be only one owner for the resulting requirement. The owner will need to be agreed, and the stakeholder that is not now owning the requirements will also need to be noted as being interested.
28	B	Topic 6 (6.4)	Slicing refers to the process of delivering requirements in an ongoing, iterative manner – in subsets – rather than en masse.
29	C	Topic 6 (6.5)	Prioritisation will use the rules to assist the prioritisation, but will not define any.
30	A	Topic 6 (6.6)	The requirement is not testable. In order for a requirement to be considered testable, there must be agreed criteria to demonstrate or check that it has been delivered.
31	C	Topic 7 (7.1)	This is the objective of user role analysis. The other options may result in having done the analysis and may play a part in the solutions delivery, but are not the purpose of user role analysis.

32	A, B and D	Topic 8 (8.1)	Traceability is vertical, back to the business objectives'; or horizontal, tracing it's journey from origin to resolution. How a requirement was validated will be recorded as a relevant document, and the owner changing over time will be change management, but not traceability.
33	B, C and D	Topic 8 (8.3)	The six elements include origin and ownership, change control and configuration management.
34	D	Topic 8 (8.2)	Requirements management focusses on the requirements themselves and ensures that they are in a known state and are able to be used in the correct manner for the purpose that is required.
35	A	Topic 8 (8.4)	The stages of the change control process are; document the proposed change, analyse the change, consult the stakeholders, decide on the change. Without a documented expression of the change required and the rationale for it, any process attempting to manage the change will have to contend with different expectations and beliefs on the change and it's likely impact.
36	A	Topic 8 (8.5)	The number 3 indicates successful validation passes. The 4 indicates the number of draft versions to the one that last passed validation.
37	D and E	Topic 5 (5.4)	Agile validation is done when the product backlog is initiated, and the backlog is established; and when maintaining the backlog until they are deemed 'ready' to progress to development. The other options are all after the development has started and validation must be completed by then.
38	C	Topic 7 (7.2)	The goal for the persona would describe why the persona is interacting with the organisation, so should contain their reason for engaging with this customer journey. The stages and touchpoints will not indicate intent, as they will be the same no matter what the purpose. Potential opportunities for improvement are internal to the organisation and how they may improve.
39	A	Topic 1 (1.3)	The project approach must be established before RE activity can commence, as this will impact the tasks undertaken.
40	D	Topic 6 (6.3)	Calculating the rate of interest and the monthly payment are two separate requirements.

End of Paper